The parabola is open left, its focus is

$$F(-a,o), F(-3,0)$$

Its vertex is $V(h,k) = V(0,0)$

The equation of the directrix is $x = a$

i.e., $x = 3$

Its axis is x – axis whose equation is $y = 0$.

Length of the latus rectum = $4a = 12$

Example 3.33

Show that the demand function $x = 10p - 20 - p^2$ is a parabola and price is maximum at its vertex.

Solution

$$\begin{aligned} x &= 10p - 20 - p^2 \\ &= -p^2 - 20 + 10p + 5 - 5 \\ x - 5 &= -p^2 + 10p - 25 \\ &= -\left(p^2 - 10p + 25\right) \\ &= -(p-5)^2 \end{aligned}$$

Put $X = x - 5$ and $P = p - 5$

$X = -P^2$ i.e $P^2 = -X$

It is a parabola open downward.

$\therefore$ At the vertex $p = 5$ when $x = 5$. i.e., price is maximum at the vertex.

Example 3.34

Find the axis, vertex, focus, equation of directrix and the length of latus rectum for the parabola $x^2 + 6x - 4y + 21 = 0$

Solution

$$\begin{aligned} 4y &= x^2 + 6x + 21 \\ 4y &= (x^2 + 6x + 9) + 12 \\ 4y - 12 &= (x+3)^2 \\ (x+3)^2 &= 4(y-3) \end{aligned}$$

$X = x + 3, Y = y - 3$

$x = X - 3$ and $y = Y + 3$

$$X^2 = 4Y$$

Comparing with $X^2 = 4aY, 4a = 4$

$$a = 1$$

	Referred to (X, Y)	Referred to (x, y) $x = X - 3$, $y = Y + 3$
Axis $y = 0$	$Y = 0$	$y = 3$
Vertex $V(0,0)$	$V(0,0)$	$V(-3,3)$
Focus $F(0, a)$	$F(0, 1)$	$F(-3, 4)$
Equation of directrix $(y=-a)$	$Y = -1$	$y = 2$
Length of Latus rectum $(4a)$	$4(1) = 4$	4

Example 3.35

The supply of a commodity is related to the price by the relation $x = \sqrt{5P - 15}$. Show that the supply curve is a parabola.

Solution

The supply price relation is given by

$$\begin{aligned} x^2 &= 5p - 15 \\ &= 5(p - 3) \end{aligned}$$

$\Rightarrow \quad X^2 = 4aP$ where $X = x$ and $P = p - 3$

$\therefore$ the supply curve is a parabola whose vertex is $(X = 0, P = 0)$

i.e., The supply curve is a parabola whose vertex is (0, 3)

Exercise 3.6

1. Find the equation of the parabola whose focus is the point $F(-1, -2)$ and the directrix is the line $4x - 3y + 2 = 0$.

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